BREAST CANCER PREDICTION

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Signature of the Student

CERTIFICATE

This is to certify Subject Specific Project entitled **"BREAST CANCER PREDICTION"** has been carried out by **Itishri Jena (2301298142)** and **Sonalita Behera(2301298233)** completed under my guidance and the project meets the academic requirement of the subject *Computer network* (BTCS-T-PC_601)

Signature of the guide

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ABSTRACT

Breast cancer is one of the most common diseases among women worldwide. Early detection of breast cancer can help in proper treatment and can save lives. This project focuses on predicting breast cancer using machine learning techniques. The system uses a dataset containing medical information of patients and applies classification algorithms to predict whether the tumor is malignant or benign. The model is trained using Python and machine learning libraries. The result shows that machine learning can help doctors in fast and accurate prediction of breast cancer.

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INTRODUCTION

In Breast cancer is a serious health problem and one of the leading causes of death among women. Detecting breast cancer at an early stage is very important for successful treatment. Traditional diagnosis methods take time and sometimes may give incorrect results.

Machine Learning is a modern technique that can help in predicting diseases using past data. In this project, a breast cancer prediction system is developed using machine learning algorithms. The system analyzes patient data and predicts whether the cancer is present or not.

This project helps in:

- Fast diagnosis
- Accurate prediction
- Supporting doctors in decision making

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LITERATURE SURVEY

Many researchers have worked on breast cancer prediction using machine learning and artificial intelligence. Early detection of breast cancer is very important because it increases the chances of successful treatment. Traditional diagnosis methods sometimes take more time and may not always give accurate results, so researchers started using computer-based prediction systems. Several studies used the **Wisconsin Breast Cancer Dataset**, which is one of the most popular datasets for testing prediction models. Different machine learning algorithms such as Logistic Regression, Support Vector Machine (SVM), Decision Tree, Random Forest, and K-Nearest Neighbor (KNN) have been applied to classify tumors as benign or malignant. Many experiments showed that these algorithms can give high accuracy in detecting breast cancer. In one study, eight different algorithms were compared, including Logistic Regression, Random Forest, KNN, Decision Tree, AdaBoost, SVM, Gradient Boosting, and Naive Bayes. The result showed that Logistic Regression achieved accuracy above 99%, proving that machine learning models can be very effective for medical prediction. Another research compared SVM, Logistic Regression, and KNN for breast cancer detection. The study found that Support Vector Machine often gives the best performance in terms of accuracy, sensitivity, and specificity, which makes it suitable for medical diagnosis systems. Researchers have also used deep learning and neural networks for detecting cancer from medical images such as mammograms. These methods can extract important features automatically and improve prediction accuracy, but they require more data and higher computing power. Recent studies show that machine learning models can help doctors by providing fast and accurate prediction, reducing human error, and supporting decision making. Therefore, breast cancer prediction using machine learning is becoming an important research area in healthcare technology.

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PROBLEM STATEMENT

Breast cancer is one of the most common and dangerous diseases among women all over the world. Early detection of breast cancer is very important to reduce the death rate and to provide proper treatment at the right time. Traditional methods of diagnosis such as manual testing and medical examination may take more time and sometimes may not give accurate results. Doctors have to analyze many medical reports, which makes the process slow and difficult.

Because of these problems, there is a need for a computerized system that can help in predicting breast cancer quickly and accurately. Machine learning techniques can be used to analyze medical data and detect patterns that are difficult to find manually. By using past patient data, the system can learn and predict whether the tumor is **benign (non-cancerous)** or **malignant (cancerous)**.

The main problem is to design and develop a breast cancer prediction system that:

- Takes patient medical data as input
- Uses machine learning algorithms for analysis
- Predicts the presence of breast cancer
- Provides fast and accurate results
- Helps doctors in decision making

The goal of this project is to build a reliable prediction model using Python and machine learning that can assist in early detection of breast cancer and improve healthcare services.

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SOURCE CODE

```
def models(X_train,Y_train):
    #logistic regression
    from sklearn.linear_model import LogisticRegression
    log=LogisticRegression(random_state=0)
    log.fit(X_train,Y_train)


    #Decision Tree
    from sklearn.tree import DecisionTreeClassifier
    tree=DecisionTreeClassifier(random_state=0,criterion="entropy")
    tree.fit(X_train,Y_train)


    #Random forest
    from sklearn.ensemble import RandomForestClassifier
    forest=RandomForestClassifier(random_state=0,criterion="entropy",n_estimators=10)
    forest.fit(X_train,Y_train)


    print('[0]logistic regression accuracy:',log.score(X_train,Y_train))
    print('[1]Decision tree accuracy:',tree.score(X_train,Y_train))
    print('[2]Random forest accuracy:',forest.score(X_train,Y_train))


    return log,tree,forest
```

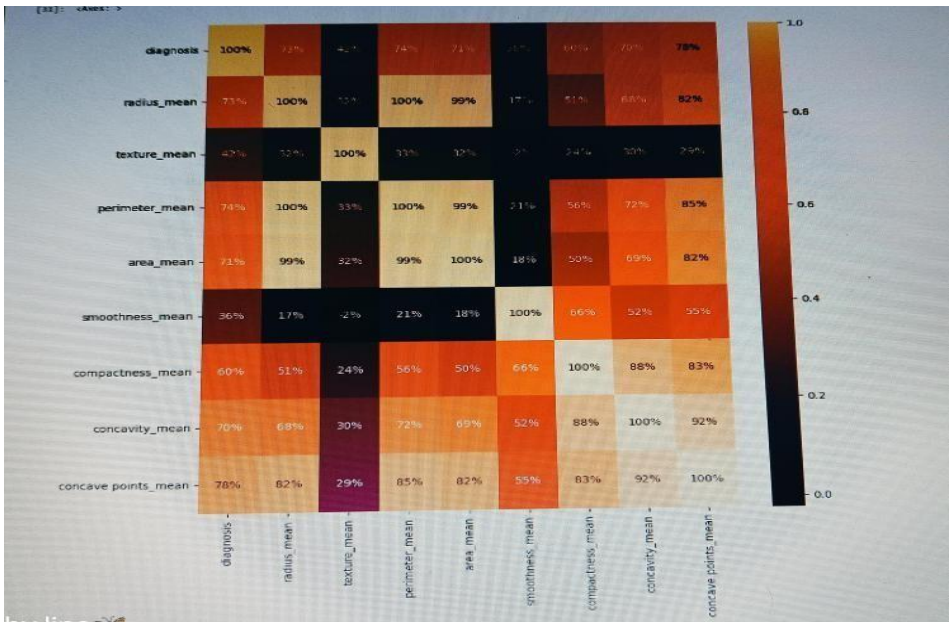
```
# testing the models/result

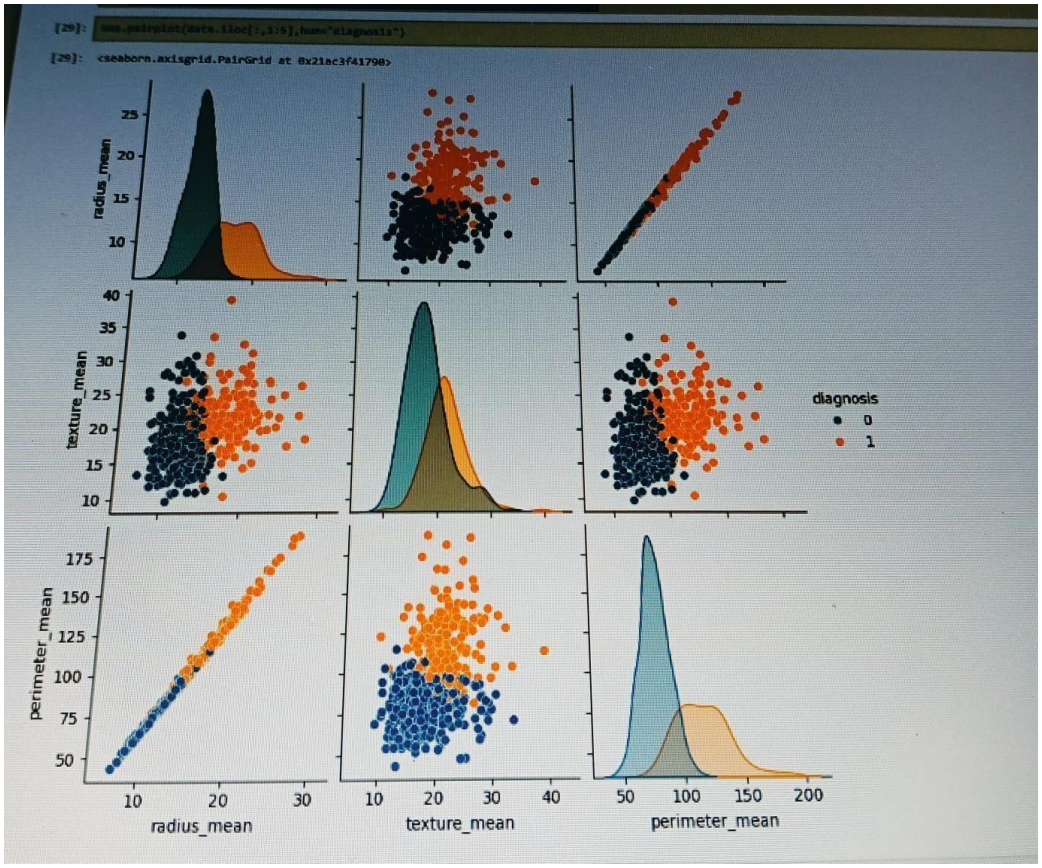
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report

for i in range(len(model)):
    print("Model", i)
    print(classification_report(Y_test, model[i].predict(X_test)))
```

```
# visualize the correlation
plt.figure(figsize=(10,10))
sns.heatmap(data.iloc[:,1:10].corr(),annot=True,fmt=".0%")
```

OUTPUT


diagnosis
radius_mean
texture_mean
perimeter_mean
area_mean
smoothness_mean
compactness_mean
concavity_mean
concave points_mean
1.0
0.8
0.6
0.4
0.2
0.0


[29]:
[29]: <seaborn.axisgrid.PairGrid at 0x21ac3f41798>
radius_mean
texture_mean
perimeter_mean
diagnosis
0
1
radius_mean
texture_mean
perimeter_mean

FUTURE INHANCEMENT

The current breast cancer prediction system uses basic machine learning algorithms to predict whether the tumor is benign or malignant. Although the system gives good accuracy, it can be improved in many ways in the future to make it more useful, reliable, and suitable for real-world medical applications.In future, more advanced **machine learning and deep learning algorithms** can be used to increase prediction accuracy. Techniques such as Neural Networks, Convolutional Neural Networks (CNN), and Artificial Intelligence can help in detecting cancer more precisely, especially when working with large medical datasets and images.The system can also be improved by using **larger and real hospital datasets** instead of small sample datasets. More data will help the model learn better patterns and give more accurate results. Connecting the system with hospital databases will make the prediction process faster and automatic.Another enhancement is to develop a **Graphical User Interface (GUI)** or **web application** so that doctors and medical staff can easily use the system without knowing programming. A simple interface with input fields and result display will make the system user-friendly.In future, the project can also include **image-based cancer detection** using mammogram or scan images. Image processing and deep learning can help in detecting cancer at a very early stage.The system can also be converted into a **real-time prediction system** that can be used in hospitals, clinics, and diagnostic centers. With cloud technology, the system can be accessed from anywhere.Therefore, with these future enhancements, the breast cancer prediction system can become more powerful, accurate, and useful in the medical field and can help doctors in saving more lives.

CONCLUSION

Breast cancer is one of the most serious diseases affecting women worldwide, and early detection is very important for successful treatment. In this project, a breast cancer prediction system was developed using machine learning techniques. The system uses medical data to predict whether the tumor is benign or malignant. By applying machine learning algorithms, the prediction process becomes faster and more accurate compared to manual diagnosis.

The project shows that machine learning can be very helpful in the medical field. The model trained on the dataset gives high accuracy and can assist doctors in making better decisions. This system reduces human effort, saves time, and helps in early detection of breast cancer, which can increase the chances of survival.

Although the current system works well, it can be improved in the future by using advanced algorithms, larger datasets, and real-time applications. With further development, this type of system can be used in hospitals and diagnostic centers to support doctors and improve healthcare services.

In conclusion, the Breast Cancer Prediction system demonstrates that machine learning is a powerful tool for disease prediction and can play an important role in saving human lives through early and accurate detection.

REFERENCE

A. UCI Machine Learning Repository – Breast Cancer Dataset

B. Python Machine Learning – Sebastian Raschka

C. Scikit-learn Documentation

D. Research papers on Breast Cancer Prediction

E. IEEE papers on Machine Learning in Healthcare